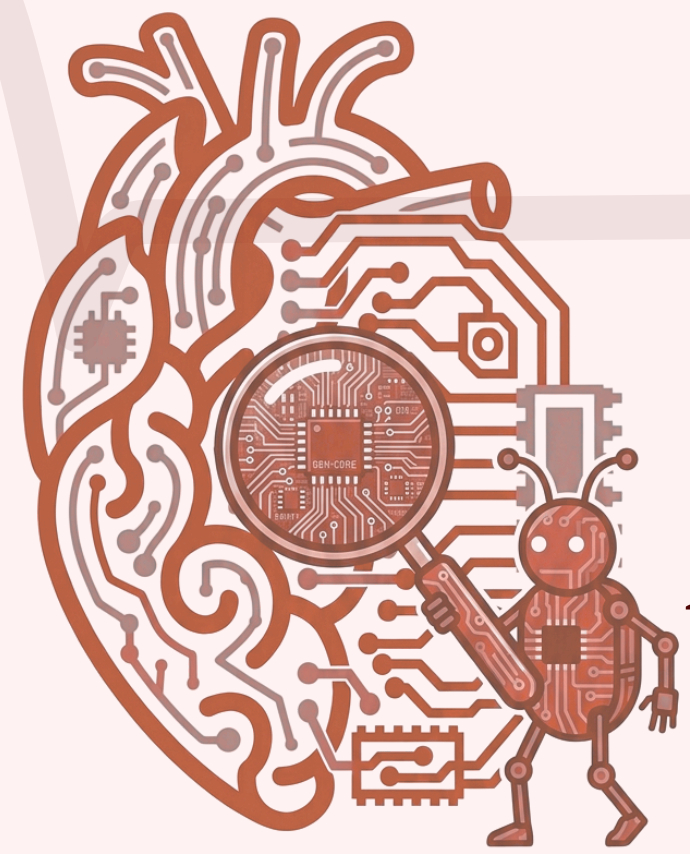


CARDIOVASCULAR DISEASE DETECTION USING DEEP LEARNING TECHNIQUES



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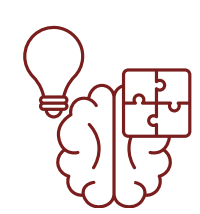
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IT DEPARTMENT

BACKGROUND



Abstract

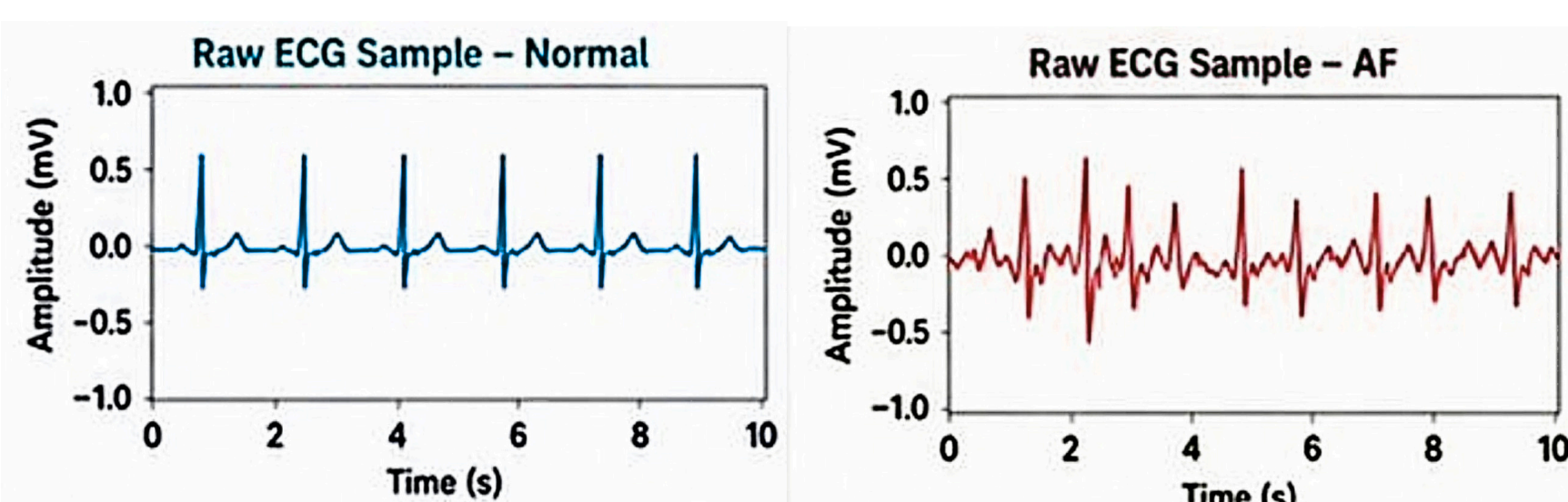
Cardiovascular diseases (CVDs) are among the most common and serious health conditions worldwide. This project presents a lightweight deep-learning framework for atrial fibrillation (AF) detection from single-lead ECG signals using the PhysioNet Challenge 2017 dataset. The proposed pipeline includes preprocessing, record-level splitting, overlapping window segmentation, and training-set balancing. A lightweight 1D-CNN with residual learning and squeeze-and-excitation attention is used to classify Normal and AF rhythms. The proposed model contains approximately 50,000 trainable parameters, enabling efficient learning with low computational complexity.



Problem statement

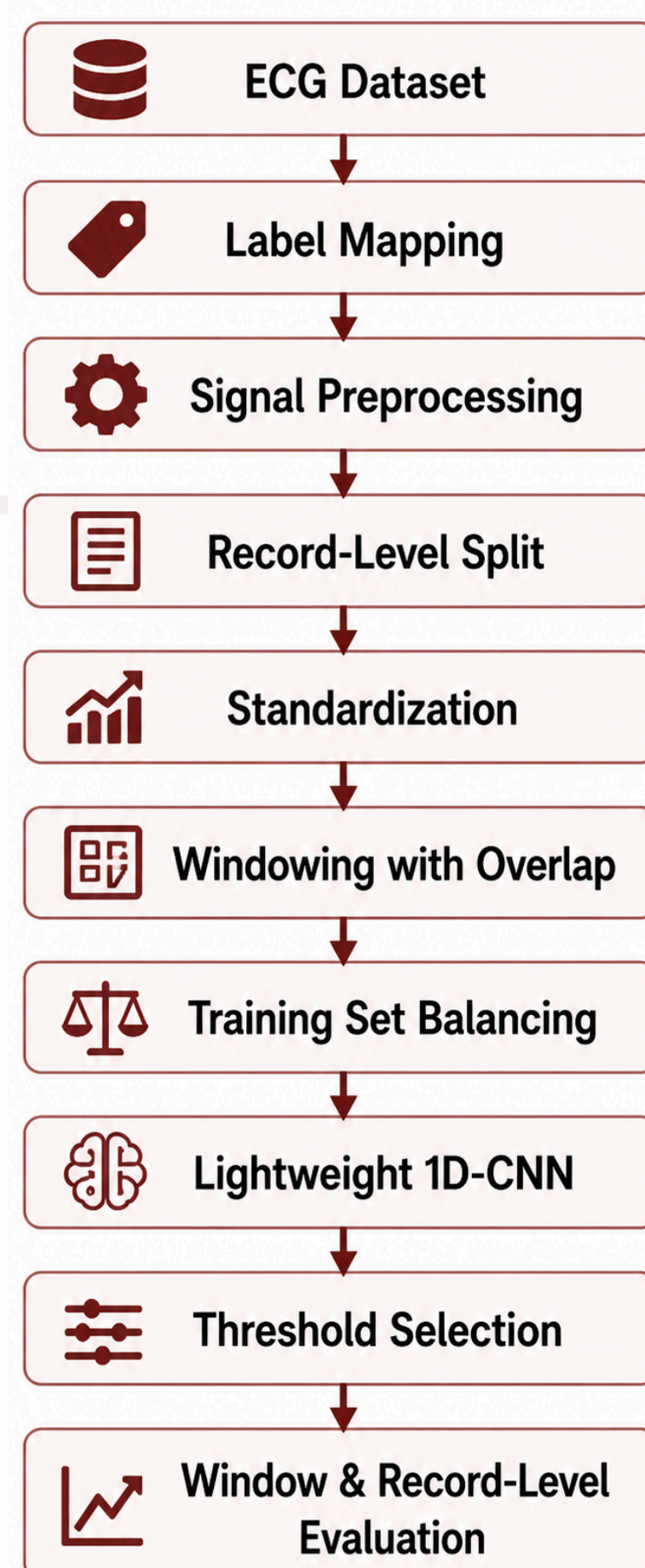
Atrial fibrillation is a common cardiac arrhythmia associated with serious complications such as stroke and heart failure. Manual ECG interpretation can be time-consuming, while several previous deep-learning approaches rely on complex architectures with high computational cost. In addition, ECG data often suffer from class imbalance and rhythm variability, making automated AF detection challenging. Therefore, an accurate and lightweight deep-learning framework is needed for reliable AF detection from ECG signals.

Raw ECG signals

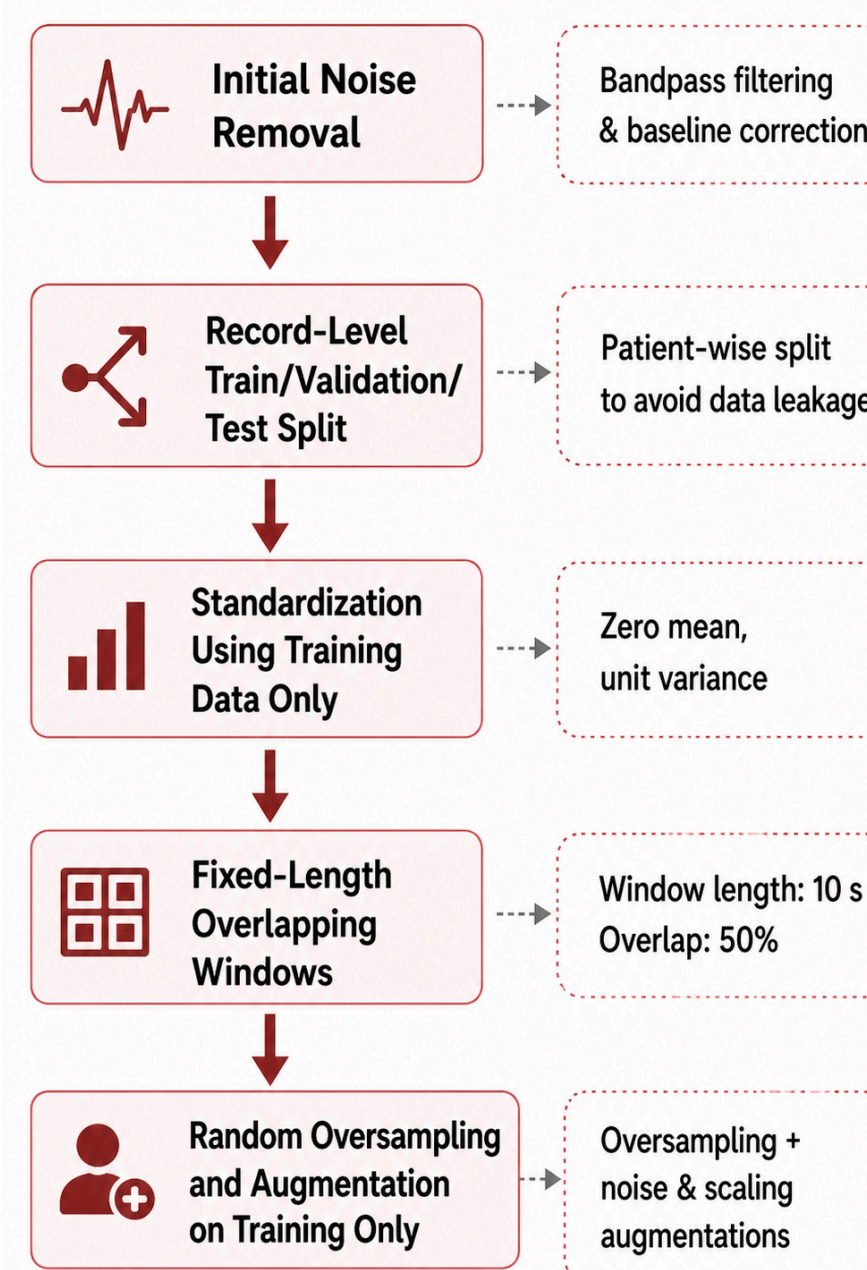


METHODOLOGY

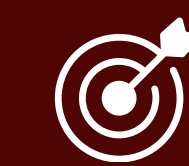
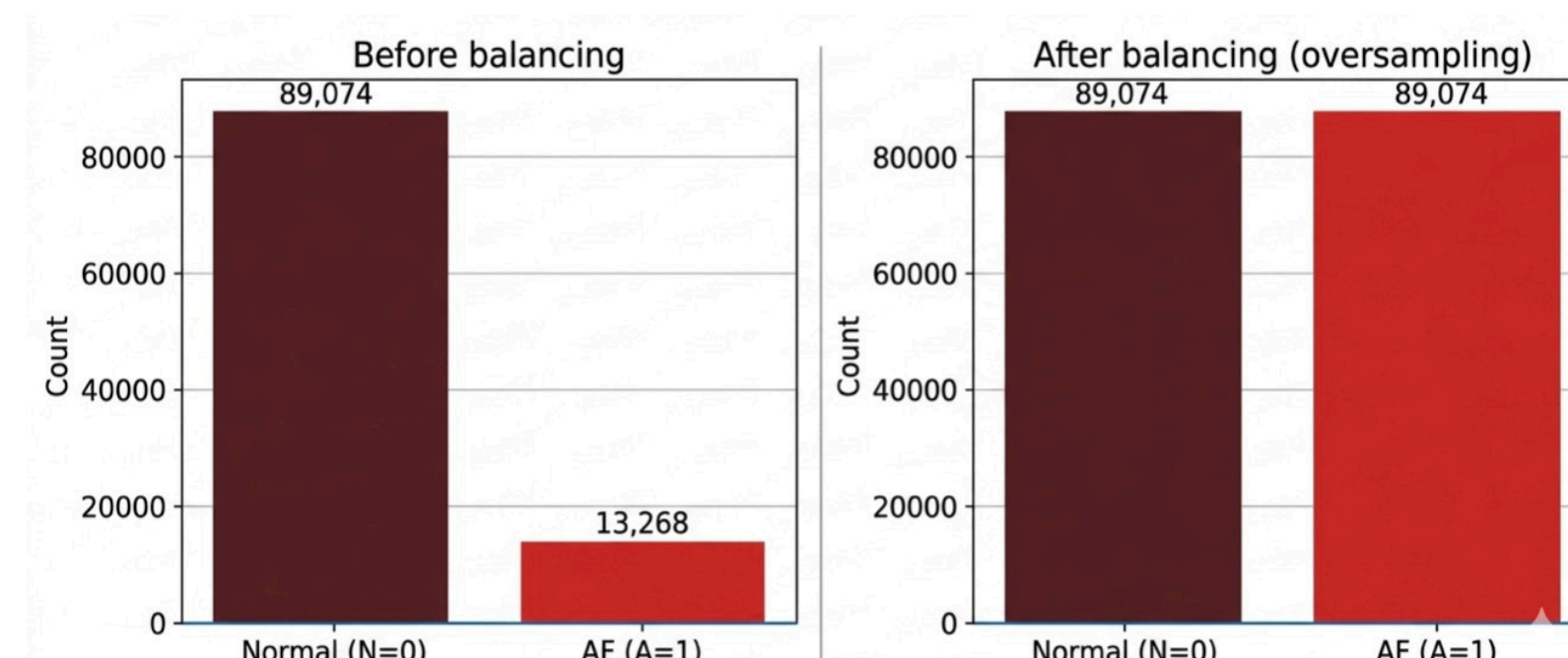
1. Methodology Pipeline



2. Data Preparation

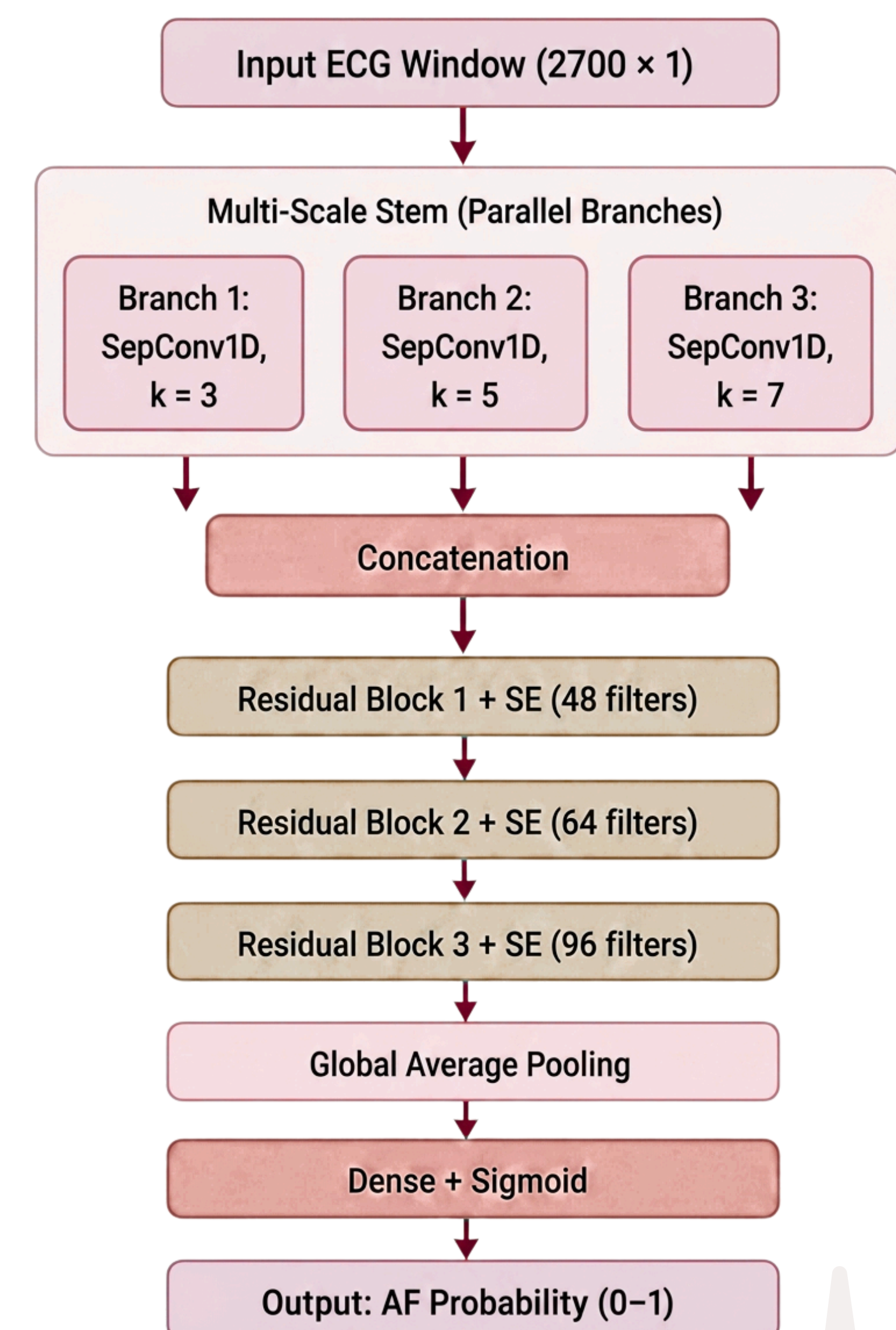


3. Training Class Distribution

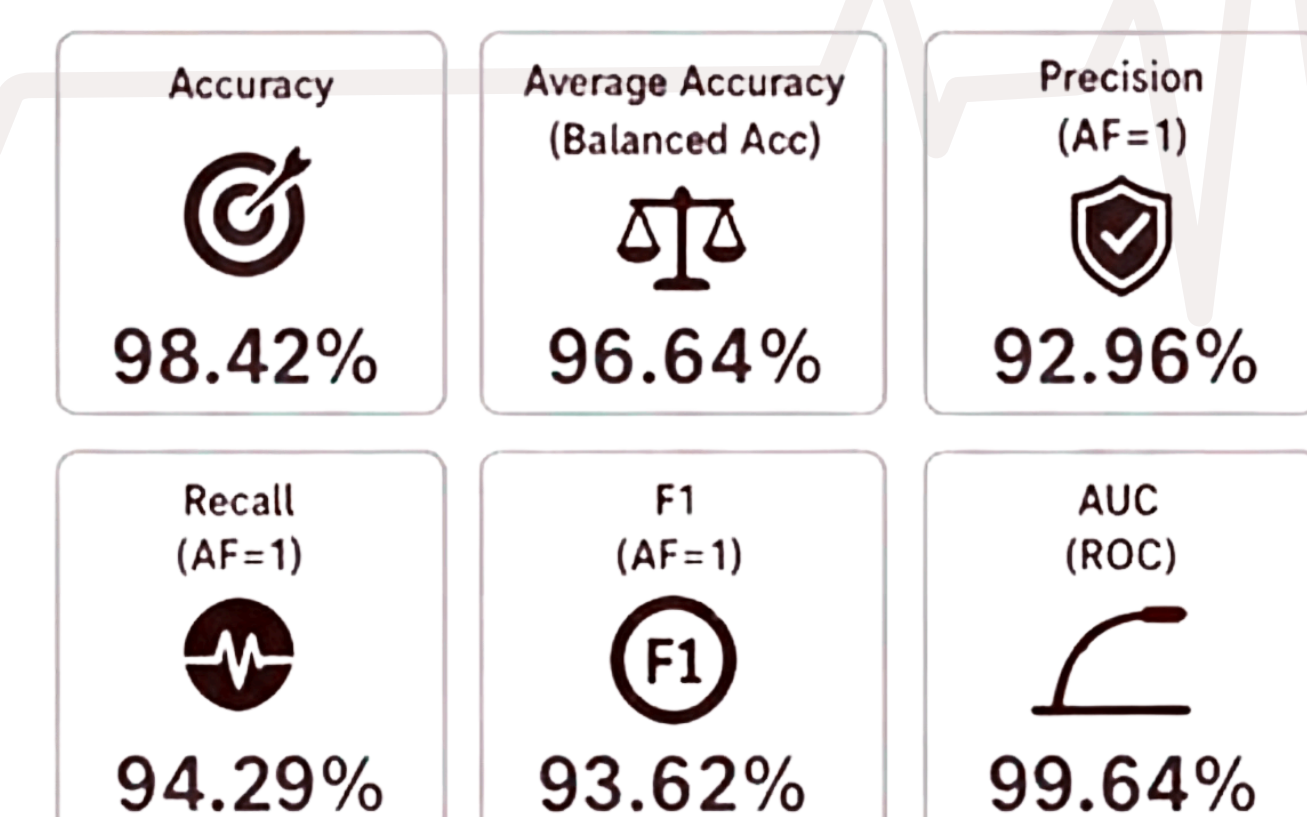


MODLE & RESULTS

1. Proposed Model



2. Results

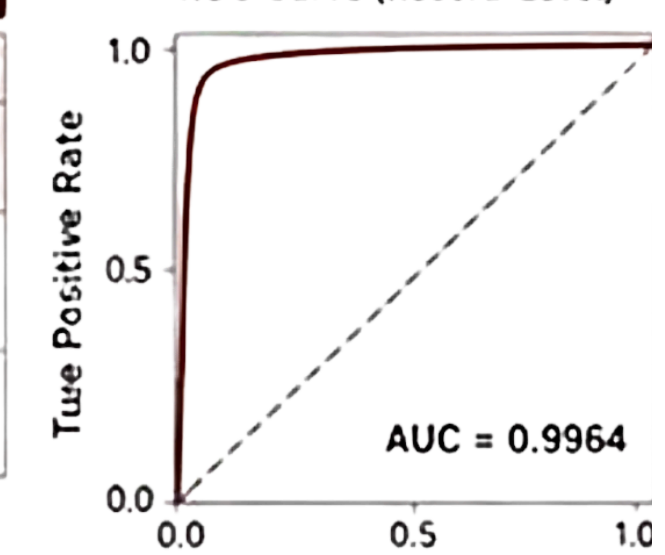


Threshold = 0.59

Confusion Matrix (Record-Level)

		Predicted	
		Normal (N=0)	AF (A=1)
Actual	Normal (N=0)	495	5
	AF (A=1)	4	66
TN = 495 FP = 5 FN = 4 TP = 66			

ROC Curve (Record-Level)



CONCLUSION

- The proposed framework achieves strong AF detection performance.
- Record-level prediction improves practical clinical interpretation.
- The lightweight architecture supports efficient ECG classification.
- Oversampling helps address severe training-data imbalance.



FUTUER WORK

- extend to multiclass ECG classification.
- evaluate on additional dataset.
- explore wearable and real-time deployment.
- further improve robustness and generalization.

